Generation Is Not Color: Resolving the Symmetric Embedding of Algebraic R-Flux in Gresnigt's Three-Generation Construction

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Abstract. Gresnigt's construction of three Standard Model fermion generations from the complexified sedenions produces three octonion subalgebras $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$, related to the physical generation structure by an order-three automorphism ψ. A companion note left open whether Günaydin–Lüst–Malek's octonion-derived algebraic R-flux algebra coincides canonically with one specific $\mathbb{O}_i$, after finding that existence-based isomorphism tests are structurally incapable of answering this, since they hold uniformly for all three regardless of any special relationship between the $\mathbb{O}_i$. We show, by direct computation, that ψ itself does *not* relate the three subalgebras to one another — it stabilizes each one individually, generating the three fermion generations entirely *within* a fixed $\mathbb{O}_i$. The symmetry actually responsible for the uniform embedding is a separate, independently verified automorphism drawn from the other factor of $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$, which exactly permutes $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ as whole subalgebras. Composing this automorphism with a known embedding of GLM's algebra into $\mathbb{O}_1$ reproduces, exactly, embeddings into $\mathbb{O}_2$ and $\mathbb{O}_3$ found by entirely independent search. Generation and color are orthogonal symmetries of the construction, and it is specifically the color symmetry — not the physically meaningful generation symmetry — that explains why algebraic R-flux embeds without preference among the three generations.

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1. Background and the question left open

A companion note poses the question of whether $\mathrm{Im}(\mathbb{O})$, the seven-dimensional Malcev algebra underlying Günaydin, Lüst & Malek's algebraic R-flux construction (arXiv:1607.06474), coincides canonically with one specific member of Gresnigt's three sedenion-derived octonion subalgebras (Gresnigt, *Eur. Phys. J. C* 79, 890, 2019), rather than merely being abstractly isomorphic to it, which Hurwitz's theorem guarantees trivially for any two octonion algebras. That note establishes, by exhaustive computational search, that existence-based tests — does *some* structure-preserving map exist, subject to natural anchor constraints — cannot distinguish this question from the trivial reading: compatible embeddings were found into all three subalgebras, in large, uniform numbers, regardless of which structurally motivated anchor points were fixed. This was traced to the octonion automorphism group's own rich transitivity, not to any specific relationship between $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ — leaving open whether the uniform embedding reflects a genuine, checkable symmetry of the construction, or is simply an artifact too weak to carry any meaning.

2. An initial hypothesis, tested and found false

Gresnigt's construction carries an explicit order-three automorphism ψ of the full sedenion algebra $\mathbb{S}$, acting on each pair $(e_i, e_{i+8}), i = 1, \ldots, 7$, as a rotation by $2\pi/3$:

$$\psi(e_i) = -\tfrac{1}{2}e_i - \tfrac{\sqrt{3}}{2}e_{i+8}, \qquad \psi(e_{i+8}) = \tfrac{\sqrt{3}}{2}e_i - \tfrac{1}{2}e_{i+8}, \qquad \psi(e_0) = e_0,\ \psi(e_8) = e_8.$$

This is the automorphism generating Gresnigt's three fermion generations from a single set of ladder operators. Given three color-labeled octonion subalgebras exist in the construction, it is natural to ask whether ψ is also what relates them to one another.

Lemma 1 (Stabilization). ψ does not relate $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ to one another; it maps each one entirely into itself.

Verification. ψ was applied to each of $\mathbb{O}_1$'s eight generators $\{e_0, e_1, e_4, e_5, e_8, e_9, e_{12}, e_{13}\}$ and the result checked for membership in the spans of $\mathbb{O}_1, \mathbb{O}_2$, and $\mathbb{O}_3$. Every image lies entirely within $\mathbb{O}_1$'s own span; none has any component along $\mathbb{O}_2$'s or $\mathbb{O}_3$'s non-shared generators. This follows directly from ψ's structure: it rotates each (e_i, e_{i+8}) pair independently and identically, and since $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ are distinguished precisely by *which* pairs they contain, no such rotation can carry one subalgebra's pairs into another's.

3. Confinement of the generations

Lemma 2. The ladder operators defining Gresnigt's three generations, A_i, $B_i = \psi(A_i)$, $C_i = \psi^2(A_i)$, are built entirely from a single $\mathbb{O}_i$'s own generators.

Verification. For $i = 1$: $A_1^\dagger, B_1^\dagger, C_1^\dagger$ are linear combinations of exactly $\{e_1, e_5, e_9, e_{13}\} \subset \mathbb{O}_1$. The linear dependence $A_i^\dagger + B_i^\dagger + C_i^\dagger = 0$, stated in Gresnigt's construction, was independently confirmed to machine precision for $i = 1, 2, 3$, using the explicit ψ-image coefficients $a = (\sqrt{3} - 1)/2, b = (-\sqrt{3} - 1)/2$.

Together, Lemmas 1 and 2 establish that the generation index (which of A, B, C) and the color index (which of $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$) are structurally orthogonal in Gresnigt's own construction — not two names for the same threefold symmetry.

4. The correct mechanism: a verified color automorphism

Since $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ (Wilmot, arXiv:2512.07210), with ψ generating the S_3 factor just shown to stabilize each $\mathbb{O}_i$, any symmetry relating $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ must be sought in the G_2 factor instead.

Lemma 3 (Color automorphism). There exist automorphisms of the full sedenion algebra, distinct from ψ, exactly permuting $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$.

Construction. Searching the (independently verified, order-1344) automorphism group of the bare octonion sub-block $e_1, \ldots, e_7$ for a signed permutation sending $\{1, 5\} \to \{2, 6\}$ while fixing 4 (the shared generator distinguishing $\mathbb{O}_1$ from $\mathbb{O}_2$ within the non-central directions) produces a candidate; extending it to the full 16-dimensional space via $e_{i+8} \mapsto s_i\, e_{\pi(i)+8}$, matching the action on $e_i \mapsto s_i\, e_{\pi(i)}$, and checking the extension directly against all 240 ordered pairs of sedenion basis products (not assumed from the extension rule) confirms it is a genuine automorphism ρ of $\mathbb{S}$. Applied to all eight of $\mathbb{O}_1$'s generators, ρ reproduces $\mathbb{O}_2$'s generator set exactly. An analogous, independently constructed and verified automorphism ρ' exactly maps $\mathbb{O}_1$ onto $\mathbb{O}_3$.

5. The composition theorem

Theorem. Let $f : \mathrm{Im}(\mathbb{O})_{\mathrm{GLM}} \to \mathbb{O}_1$ be an embedding found by the direct search of the companion note. Then $\rho \circ f$ is exactly one of the embeddings $\mathrm{Im}(\mathbb{O})_{\mathrm{GLM}} \to \mathbb{O}_2$ found by an entirely independent direct search into $\mathbb{O}_2$; likewise $\rho' \circ f$ exactly reproduces an independently-found embedding into $\mathbb{O}_3$.

Verification. Both compositions were computed explicitly and compared, generator by generator and sign by sign, against the full list of embeddings each independent search returned. Both match exactly, not merely up to some further unaccounted symmetry.

This is the substantive result: the uniform existence of embeddings into all three $\mathbb{O}_i$, previously shown to be too weak a fact to carry meaning on its own, is here shown to be *properly explained* — not just accompanied — by a specific, verified, non-trivial automorphism of the ambient sedenion structure. The diagram GLM$\to \mathbb{O}_1 \to \mathbb{O}_2$ (via ρ) commutes with the direct search, exactly, not approximately.

6. What this resolves

The original question — does GLM's algebra coincide canonically with *one* specific $\mathbb{O}_i$ — is answered here, precisely, in the negative, and for a stronger reason than mere existence-symmetry. It is not merely that embeddings happen to exist into all three; it is that a genuine, checked symmetry of the full construction carries any one embedding onto the others exactly. The three embeddings are not independent facts each separately requiring explanation — they are the same fact, related by ρ and ρ', the color automorphism. There is no privileged $\mathbb{O}_i$ for algebraic R-flux to canonically prefer, and this is now established rather than merely undecidable.

It is worth being precise about what this does *not* show: it does not show that GLM's construction is related to *Gresnigt's own generation structure* in any physically meaningful way — ψ, the symmetry actually tied to the physics of three fermion generations, plays no role in this result at all (Lemma 1). The resolution concerns the algebra's internal color symmetry, not a statement about which fermion generation, if any, algebraic R-flux might be associated with.

7. Scope

This was verified for the pair $(\mathbb{O}_1, \mathbb{O}_2)$, the pair $(\mathbb{O}_1, \mathbb{O}_3)$, and — via the composition $\sigma = \rho' \circ \rho$, independently confirmed to be a genuine automorphism of $\mathbb{S}$ satisfying $\sigma(\mathbb{O}_2) = \mathbb{O}_3$ exactly — the pair $(\mathbb{O}_2, \mathbb{O}_3)$ as well, closing the triad completely: composing σ with a known $\mathbb{O}_2$-embedding reproduces a known $\mathbb{O}_3$-embedding exactly. It was not checked whether *every* embedding into $\mathbb{O}_1$ composes correctly under ρ, ρ', σ, only the specific representatives used throughout.

References

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